

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

Table of Contents

Time-Domain Astrophysics

Academic Year 2025-2026

Lecture list:

TIME-DOMAIN ASTROPHYSICS

Stefano Covino

INAF / Brera Astronomical Observatory



Time-Domain Astrophysics

Academic Year 2025-2026

.....

Lecture list:

1. Lecture: Introduction	notebook	html
2. Lecture: Statistics Reminder	notebook	html
3. Lecture: Spectral Analysis	notebook	html
4. Science case: Sunspot number	notebook	html
5. Science case: X-ray binaries	notebook	html
6. Lecture: Irregular sampling	notebook	html
7. Science case: Variable stars	notebook	html
8. Lecture: Wavelet Analysis	notebook	html
9. Science case: Climate data	notebook	html
10. Lecture: Time Domain analysis	notebook	html
11. Science case: AGN and Blazars	notebook	html
12. Lecture: Time of Arrival	notebook	html
13. Science case: FRBs	notebook	html
14. Lecture: Non Parametric Analysis	notebook	html
15. Science case: GRBs	notebook	html
16. Lecture: Singular Spectrum Analysis	notebook	html
17. Science case: Motion sensor data	notebook	html
18. Lecture: Gaussian Processes	notebook	html
19. Science case: CO ₂ content in atmosphere	notebook	html
20. Lecture: Interacting with an AI	notebook	html
21. Lecture: Astrostatistics Future	notebook	html

Copyright

This notebook is provided as [Open Educational Resource](#). Feel free to use the notebook for your own purposes. The text is licensed under [Creative Commons Attribution 4.0](#), the code of the examples, unless obtained from other properly quoted sources, under the [MIT license](#). Please attribute the work as follows: *Stefano Covino, Time Domain Astrophysics - Lecture notes featuring computational examples, 2026*.

Notebook v1.0.1 - 17 May 2026